University of Colombo, Sri Lanka

UCSC *University of Colombo School of Computing*

Bachelor of Science in Computer Science

First Year Examination — Semester II— UCSC AY22 [held in March/April 2026]

SCS1312 — Operating System Concepts

(2 Hours)

Answer All Questions

Number of Pages = 12 | Number of Questions = 4

45

To be completed by the candidate

Index Number | | | | | | | | |

Important Instructions to candidates:

- Students should answer in the medium of English language only using the space provided in this question paper.
- Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- Write your index number CLEARLY on each and every page of this Question paper.
- This paper consists of **4** questions in **12** pages (including the Cover Page).
- Answer ALL questions.
- Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are not allowed.
- Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate

To be completed by the examiners

1	
2	
3	
4	
Total	

Index Number | | | | | | | | |

1. (a). The following questions are related to the Mater Boot Records (MBR) of a storage device intended to be used in an IBM-compatible PC. Assume that this device is bootable.

   i. What is the content of the byte 510 of the MBR in hexadecimal notation?

   [2 marks]

   ii. What is the content of the byte 511 of the MBR in hexadecimal notation?

   [2 marks]

   iii. What is the numerical value of the last two bytes of the MBR as interpreted in an $x86$ system? Give your answer in the hexadeicmal notation and justify the answer.

   [5 marks]

   iv. What is the purpose of the byte 510 and 511?

   [2 marks]

   v. What is the content of the 64 bytes from the byte 446 to 509?

   [2 marks]

Index Number | | | | | | | | |

vi. What is the content of the first 446 bytes starting from the byte 0?

[2 marks]

(b). $x86$ family of processors have several ring levels.

i. What are the ring levels used by the Linux operating system on an $x86$ based system? Indicate the use of each ring level in Linux.

[3 marks]

ii. What is the ring level of the processor right after a page fault?

[2 marks]

iii. What is the purpose of the `int 80` instruction with reference to its use by Linux processes?

[3 marks]

iv. What is the ring level of a processor right after executing `int 80` instruction by the process running in a Linux system?

[2 marks]

Index Number | | | | | | | | |

2. (a). i. What is the output of the following C programme? Assume that all `fork()` calls completed successfully.

```
int main()
{         int x;
          x=2;
          fork();
          x=x-1;
          fork();
          x=x-1;
          printf("%d\n", x);
}
```

[4 marks]

ii. What is the output of the following C programme? Assume that all `fork()` calls completed successfully.

```
int main()
{
    int x=9;

    if(fork()==fork())
        printf("%d\n",x);
}
```

[4 marks]

iii. The function `int f(int x)` is to be used in a C programme running on a Linux system. `f()` should create a child process and should return `x` in the parent and `x+1` in the child. Write the function `f()`. To get the full marks `f()` should have only a single statement in the function body.

[5 marks]

Index Number | | | | | | | | |

iv. A process P_1 used `fork()` to create a child process P_2 in a 32-bit system running Linux. The logical address space of P_1 is 2^{32} bytes. At the time of calling `fork()`, P_1 has only one page in the physical memory. What is is the size of the logical address space of P_2? Justify your answer in **one sentence**.

[4 marks]

(b). A program P has the statement X. Multiple processes executes the program P concurrently. At most n processes are allowed to execute the statement X concurrently. Write a code segment of P, in C-like pseudo code, to use semaphores to ensure that at most n processes are allowed to execute the statement X concurrently.

[5 marks]

(c). Draw a process state transition diagram, clearly indicating the states and the events.

[3 marks]

Index Number | | | | | | | | |

3. (a). A research database stores climate data records. Each record contains the Timestamp, Location ID and the Temperature. Each record occupies 128 bytes. The system stores the data in files using 4 KB blocks.

i. How many records can be stored in a single disk block ?

[2 marks]

ii. The system needs to support two access patterns:

A. Scientists often read the entire file sequentially.

B. Occasionally a program retrieves the record for a specific timestamp.

Which file access method would be most appropriate for this scenario? Justify your choice and, if applicable, provide a suitable file organization or structure to support the selected access method.

[7 marks]

(b). A file system manages 10,000 disk blocks. Currently, 6200 blocks are allocated and 3800 blocks are free. The system supports two free-space management techniques: (i) Bitmap / Bit-vector and (ii) Linked free list.

i. How many records can be stored in a single disk block ?

[2 marks]

ii. Suppose the following bitmap fragment represents disk blocks:

Block number	20	21	22	23	24	25	26	27
Bitmap	1	1	0	0	1	0	0	1

Interpret the meaning of each bit in this bitmap with respect to each block number in the provided space below.

Index Number | | | | | | | | |

[4 marks]

iii. Compare the performance of bitmap and free list methods for the following operations: (i) Finding a free block, (ii) Allocating multiple contiguous blocks, (iii) Memory overhead and (iv) Disk scanning performance.

[8 marks]

iv. Explain by providing two (2) reasons why modern file systems use bitmap-based free space management.

[2 marks]

Index Number | | | | | | | | |

4. (a). A server with 2 CPU cores schedules the following processes. All arrive at time 0.

Process	CPU Burst (ms)
P1	10
P2	4
P3	6
P4	2
P5	8

i. Write down two (2) advantages when using indexed allocation method for a file system compared to linked allocation scheme.

[2 marks]

ii. Calculate the average turnaround time, assuming Shortest the Job First scheduling.

[2 marks]

iii. Explain how a shared/common **ready queue** improves load balancing in multicore scheduling.

[2 marks]

Index Number | | | | | | | | |

(b). A modern operating system uses demand paging with a two-level hierarchical page table and a Translation Lookaside Buffer (TLB). The system parameters are as follows:

- Virtual address size = 32 bits
- Page size = 4 KB
- Physical memory = 512 MB
- Page table entry size = 4 bytes
- TLB access time = 10 ns
- Main memory access time = 100 ns
- Disk page fault service time = 8 ms

Additional information:

- TLB hit ratio = 90%
- Page fault rate = 0.002

Assume that when a TLB miss occurs, the system must access the two-level page table in memory before accessing the desired data.

i. Determine the following :

A. Number of bits used for the page offset

B. Number of virtual pages

C. Number of physical frames

[3 marks]

ii. Each page table must fit in one page (4 KB). Determine:

A. Number of entries per page table

B. Bits used for first-level page directory

C. Bits used for second-level page table

[3 marks]

Index Number

iii. Effective memory access time including page faults are given in the following equation:

$$EMAT = (1 - p) \times MA + p \times PF$$

MA = Memory Access
PF = Page fault service
p = page fault rate
Compute the following memory access costs:

A. TLB hit cost
B. TLB miss cost
C. Page table access cost
D. Memory access cost
E. EMAT

[3 marks]

(c). A hypervisor runs two virtual machines VM1 and VM2 on a physical CPU using the trap-and-emulate virtualization technique. The following instructions are executed by VM1:

Instruction	**Type**
ADD R1, R2	User instruction
MOV CR3, R4	Privileged instruction
IN PORT1	I/O instruction
SUB R3, R5	User instruction

Assume that guest OS runs in user mode while the hypervisor runs in privileged mode.

Index Number | | | | | | | | |

i. Identify which instructions will cause a trap to the hypervisor.

[2 marks]

ii. Explain each step of the trap-and-emulate process for the privileged instruction.

[4 marks]

Index Number | | | | | | | | |

iii. Explain why trap-and-emulate works only when sensitive instructions are privileged.

[2 marks]

iv. Explain why shadow page tables cause high overhead and how modern OS eliviate this problem.

[2 marks]

****